

Model Selection — Binary Classification of Obesity Risk

Project Context

This project applies an end-to-end machine learning workflow to a structured health dataset, with the goal of predicting obesity risk based on behavioral and demographic attributes. The focus is on building a practical, decision-oriented model, aligned with applied analytics and Business Intelligence use cases, rather than purely academic classification.

[UCI Repository](https://doi.org/10.24432/C5H31Z) Estimation of Obesity Levels Based On Eating Habits and Physical Condition [Dataset]. (2019). UCI Machine Learning Repository. <https://doi.org/10.24432/C5H31Z>.

Problem Definition

The original multi-class obesity labels are reformulated into a binary classification task:

0 — No Obesity Risk (Normal Weight + Insufficient Weight)

1 — Obesity Risk (Overweight + Obesity categories)

This framing supports real-world screening scenarios, where identifying individuals at risk is more relevant than fine-grained categorical distinctions.

Model Selection Goal

The objective of this notebook is to select the most appropriate classification model using AutoSklearn2 with cross-validation.

This notebook focuses exclusively on model selection. Hyperparameter optimization and final model evaluation are performed in subsequent notebooks.

```
In [1]: from auto_sklearn2 import AutoSklearnClassifier

import pandas as pd
import numpy as np

from sklearn.model_selection import StratifiedKFold
from sklearn.metrics import roc_auc_score, f1_score, classification_report
```

Load Data

```
In [2]: import pandas as pd
from pathlib import Path
```

```
# Define project root dynamically
PROJECT_ROOT = Path.cwd()

# Path to data folder
DATA_PATH = PROJECT_ROOT / "Data"

# Load dataset
df = pd.read_csv(DATA_PATH / "ObesityDataSet_raw_and_data_synthetic.csv")

df.head()
```

Out[2]:

	Gender	Age	Height	Weight	family_history_with_overweight	FAVC	FCVC	NCP
0	Female	21.0	1.62	64.0	yes	no	2.0	3.0
1	Female	21.0	1.52	56.0	yes	no	3.0	3.0
2	Male	23.0	1.80	77.0	yes	no	2.0	3.0
3	Male	27.0	1.80	87.0	no	no	3.0	3.0
4	Male	22.0	1.78	89.8	no	no	2.0	1.0

In [3]:

```
# Binary target
df["obesity_risk"] = df["NObeyesdad"].apply(
    lambda x: 0 if x in ["Insufficient_Weight", "Normal_Weight"] else 1
)

y = df["obesity_risk"]

# Remove target original e variáveis com Leakage
X = df.drop(
    columns=[
        "NObeyesdad",
        "obesity_risk",
        "Height",
        "Weight"
    ]
)

X = X.reset_index(drop=True)
y = y.reset_index(drop=True)
```

In [4]:

```
# Identify categorical columns
categorical_columns = X.select_dtypes(include="object").columns
categorical_columns
```

Out[4]:

```
Index(['Gender', 'family_history_with_overweight', 'FAVC', 'CAEC', 'SMOKE',
       'SCC', 'CALC', 'MTRANS'],
      dtype='object')
```

In [5]:

```
# One-Hot Encoding
X_encoded = pd.get_dummies(
    X,
    columns=categorical_columns,
```

```
drop_first=True
)
```

In [6]: `X_encoded.shape`

Out[6]: (2111, 21)

```
In [7]: X = X.reset_index(drop=True)
y = y.reset_index(drop=True)

X.shape, y.shape
```

Out[7]: ((2111, 14), (2111,))

Cross-Validation Setup

```
In [8]: k = 5

cv = StratifiedKFold(
    n_splits=k,
    shuffle=True,
    random_state=42
)
```

AutoSklearn2 — Model Selection via CV

```
In [9]: cv_roc_auc = []
cv_f1 = []

for fold, (train_idx, val_idx) in enumerate(cv.split(X_encoded, y), start=1):

    X_train_cv = X_encoded.iloc[train_idx]
    X_val_cv = X_encoded.iloc[val_idx]
    y_train_cv = y.iloc[train_idx]
    y_val_cv = y.iloc[val_idx]

    automl = AutoSklearnClassifier(
        time_limit=120,
        random_state=42
    )

    automl.fit(X_train_cv, y_train_cv)

    y_val_pred = automl.predict(X_val_cv)
    y_val_proba = automl.predict_proba(X_val_cv)[:, 1]

    cv_roc_auc.append(roc_auc_score(y_val_cv, y_val_proba))
    cv_f1.append(f1_score(y_val_cv, y_val_pred))
```



```

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ValueError: Negative values in data passed to MultinomialNB (input X).
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self._count(X, Y)

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File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\utils\validation.py", line 1775, in check_non_negative

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    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\naive_bayes.p
y", line 794, in fit
    self._count(X, Y)
  File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\naive_bayes.p
y", line 921, in _count
    check_non_negative(X, "MultinomialNB (input X)")
  File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\utils\validat
ion.py", line 1775, in check_non_negative
    raise ValueError(f"Negative values in data passed to {whom}.")
ValueError: Negative values in data passed to MultinomialNB (input X).

```

Error evaluating robust_scaler + qda:
All the 5 fits failed.
It is very likely that your model is misconfigured.
You can try to debug the error by setting error_score='raise'.

Below are more details about the failures:

```

-----
4 fits failed with the following error:
Traceback (most recent call last):

```

```

File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\model_selecti
on_validation.py", line 833, in _fit_and_score
    estimator.fit(X_train, y_train, **fit_params)
File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\base.py", lin
e 1336, in wrapper
    return fit_method(estimator, *args, **kwargs)
    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\pipeline.py",
line 621, in fit
    self._final_estimator.fit(Xt, y, **last_step_params["fit"])
File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\base.py", lin
e 1336, in wrapper
    return fit_method(estimator, *args, **kwargs)
    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\discriminant_
analysis.py", line 1132, in fit
    raise linalg.LinAlgError(
numpy.linalg.LinAlgError: The covariance matrix of class 1 is not full rank. Increas
e the value of `reg_param` to reduce the collinearity.

```

```

-----
1 fits failed with the following error:
Traceback (most recent call last):
File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\model_selecti
on_validation.py", line 833, in _fit_and_score
    estimator.fit(X_train, y_train, **fit_params)
File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\base.py", lin
e 1336, in wrapper
    return fit_method(estimator, *args, **kwargs)
    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\pipeline.py",
line 621, in fit
    self._final_estimator.fit(Xt, y, **last_step_params["fit"])
File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\base.py", lin
e 1336, in wrapper
    return fit_method(estimator, *args, **kwargs)
    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\discriminant_
analysis.py", line 1132, in fit
    raise linalg.LinAlgError(
numpy.linalg.LinAlgError: The covariance matrix of class 0 is not full rank. Increas
e the value of `reg_param` to reduce the collinearity.

```

Error evaluating standard_scaler + multinomial_nb:
All the 5 fits failed.
It is very likely that your model is misconfigured.
You can try to debug the error by setting error_score='raise'.

Below are more details about the failures:

```

-----
5 fits failed with the following error:
Traceback (most recent call last):
File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\model_selecti
on_validation.py", line 833, in _fit_and_score
    estimator.fit(X_train, y_train, **fit_params)
File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\base.py", lin

```



```

File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\pipeline.py",
line 621, in fit
    self._final_estimator.fit(Xt, y, **last_step_params["fit"])
File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\base.py", line
1336, in wrapper
    return fit_method(estimator, *args, **kwargs)
    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\discriminant_
analysis.py", line 1132, in fit
    raise linalg.LinAlgError(
numpy.linalg.LinAlgError: The covariance matrix of class 0 is not full rank. Increase
the value of `reg_param` to reduce the collinearity.

```

Error evaluating robust_scaler + multinomial_nb:
All the 5 fits failed.
It is very likely that your model is misconfigured.
You can try to debug the error by setting error_score='raise'.

Below are more details about the failures:

```

-----
5 fits failed with the following error:
Traceback (most recent call last):
  File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\model_selecti
on_validation.py", line 833, in _fit_and_score
    estimator.fit(X_train, y_train, **fit_params)
  File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\base.py", line
1336, in wrapper
    return fit_method(estimator, *args, **kwargs)
    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\pipeline.py",
line 621, in fit
    self._final_estimator.fit(Xt, y, **last_step_params["fit"])
  File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\base.py", line
1336, in wrapper
    return fit_method(estimator, *args, **kwargs)
    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\naive_bayes.p
y", line 794, in fit
    self._count(X, Y)
  File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\naive_bayes.p
y", line 921, in _count
    check_non_negative(X, "MultinomialNB (input X)")
  File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\utils\validat
ion.py", line 1775, in check_non_negative
    raise ValueError(f"Negative values in data passed to {whom}.")
ValueError: Negative values in data passed to MultinomialNB (input X).

```

Error evaluating robust_scaler + qda:
All the 5 fits failed.
It is very likely that your model is misconfigured.
You can try to debug the error by setting error_score='raise'.

Below are more details about the failures:

```

-----
4 fits failed with the following error:
Traceback (most recent call last):

```

```
File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\model_selection\_validation.py", line 833, in _fit_and_score
    estimator.fit(X_train, y_train, **fit_params)
File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\base.py", line 1336, in wrapper
    return fit_method(estimator, *args, **kwargs)
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^

File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\pipeline.py", line 621, in fit
    self._final_estimator.fit(Xt, y, **last_step_params["fit"])
File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\base.py", line 1336, in wrapper
    return fit_method(estimator, *args, **kwargs)
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^

File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\discriminant_analysis.py", line 1132, in fit
    raise linalg.LinAlgError(
numpy.linalg.LinAlgError: The covariance matrix of class 1 is not full rank. Increase the value of `reg_param` to reduce the collinearity.

-----
1 fits failed with the following error:
Traceback (most recent call last):
File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\model_selection\_validation.py", line 833, in _fit_and_score
    estimator.fit(X_train, y_train, **fit_params)
File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\base.py", line 1336, in wrapper
    return fit_method(estimator, *args, **kwargs)
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^

File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\pipeline.py", line 621, in fit
    self._final_estimator.fit(Xt, y, **last_step_params["fit"])
File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\base.py", line 1336, in wrapper
    return fit_method(estimator, *args, **kwargs)
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^

File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\discriminant_analysis.py", line 1132, in fit
    raise linalg.LinAlgError(
numpy.linalg.LinAlgError: The covariance matrix of class 0 is not full rank. Increase the value of `reg_param` to reduce the collinearity.
```

Cross-Validation Summary

```
In [10]: print("\n=== Cross-Validation Summary ===")
print(f"Mean ROC-AUC: {np.mean(cv_roc_auc):.4f}")
print(f"ROC-AUC Std: {np.std(cv_roc_auc):.4f}")
print(f"Mean F1: {np.mean(cv_f1):.4f}")
```

```
=== Cross-Validation Summary ===
Mean ROC-AUC: 0.9779
ROC-AUC Std: 0.0059
Mean F1: 0.9509
```

Train Final AutoSklearn2 Model

```
In [11]: automl_final = AutoSklearnClassifier(  
        time_limit=120,  
        random_state=42  
    )  
  
    automl_final.fit(X_encoded, y)
```



```

File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\discriminant_
analysis.py", line 1132, in fit
    raise linalg.LinAlgError(
numpy.linalg.LinAlgError: The covariance matrix of class 1 is not full rank. Increas
e the value of `reg_param` to reduce the collinearity.

-----
1 fits failed with the following error:
Traceback (most recent call last):
  File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\model_selectio
on\_validation.py", line 833, in _fit_and_score
    estimator.fit(X_train, y_train, **fit_params)
  File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\base.py", lin
e 1336, in wrapper
    return fit_method(estimator, *args, **kwargs)
          ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\pipeline.py",
line 621, in fit
    self._final_estimator.fit(Xt, y, **last_step_params["fit"])
  File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\base.py", lin
e 1336, in wrapper
    return fit_method(estimator, *args, **kwargs)
          ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\discriminant_
analysis.py", line 1132, in fit
    raise linalg.LinAlgError(
numpy.linalg.LinAlgError: The covariance matrix of class 0 is not full rank. Increas
e the value of `reg_param` to reduce the collinearity.

Error evaluating minmax_scaler + qda:
All the 5 fits failed.
It is very likely that your model is misconfigured.
You can try to debug the error by setting error score='raise'.

```

Below are more details about the failures:

```
4 fits failed with the following error:
Traceback (most recent call last):
  File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\model_selection_validation.py", line 833, in _fit_and_score
    estimator.fit(X_train, y_train, **fit_params)
  File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\base.py", line 1336, in wrapper
    return fit_method(estimator, *args, **kwargs)
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\pipeline.py", line 621, in fit
    self._final_estimator.fit(Xt, y, **last_step_params["fit"])
  File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\base.py", line 1336, in wrapper
    return fit_method(estimator, *args, **kwargs)
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\discriminant_analysis.py", line 1132, in fit
    raise linalg.LinAlgError(
```

```
numpy.linalg.LinAlgError: The covariance matrix of class 1 is not full rank. Increase the value of `reg_param` to reduce the collinearity.
```

```
-----
1 fits failed with the following error:
```

```
Traceback (most recent call last):
```

```
File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\model_selection\_validation.py", line 833, in _fit_and_score
```

```
    estimator.fit(X_train, y_train, **fit_params)
```

```
File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\base.py", line 1336, in wrapper
```

```
    return fit_method(estimator, *args, **kwargs)
```

```
^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\pipeline.py", line 621, in fit
```

```
    self._final_estimator.fit(Xt, y, **last_step_params["fit"])
```

```
File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\base.py", line 1336, in wrapper
```

```
    return fit_method(estimator, *args, **kwargs)
```

```
^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\discriminant_analysis.py", line 1132, in fit
```

```
    raise linalg.LinAlgError(
```

```
numpy.linalg.LinAlgError: The covariance matrix of class 0 is not full rank. Increase the value of `reg_param` to reduce the collinearity.
```

```
Error evaluating robust_scaler + multinomial_nb:
```

```
All the 5 fits failed.
```

```
It is very likely that your model is misconfigured.
```

```
You can try to debug the error by setting error_score='raise'.
```

```
Below are more details about the failures:
```

```
-----
5 fits failed with the following error:
```

```
Traceback (most recent call last):
```

```
File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\model_selection\_validation.py", line 833, in _fit_and_score
```

```
    estimator.fit(X_train, y_train, **fit_params)
```

```
File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\base.py", line 1336, in wrapper
```

```
    return fit_method(estimator, *args, **kwargs)
```

```
^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\pipeline.py", line 621, in fit
```

```
    self._final_estimator.fit(Xt, y, **last_step_params["fit"])
```

```
File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\base.py", line 1336, in wrapper
```

```
    return fit_method(estimator, *args, **kwargs)
```

```
^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\naive_bayes.py", line 794, in fit
```

```
    self._count(X, Y)
```

```
File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\naive_bayes.py", line 921, in _count
```

```
    check_non_negative(X, "MultinomialNB (input X)")
```

```
File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\utils\validation.py", line 100, in check_non_negative
```

```
ion.py", line 1775, in check_non_negative
    raise ValueError(f"Negative values in data passed to {whom}.")
ValueError: Negative values in data passed to MultinomialNB (input X).
```

Error evaluating robust_scaler + qda:
All the 5 fits failed.
It is very likely that your model is misconfigured.
You can try to debug the error by setting error_score='raise'.

Below are more details about the failures:

```
-----
4 fits failed with the following error:
Traceback (most recent call last):
  File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\model_selecti
on_validation.py", line 833, in _fit_and_score
    estimator.fit(X_train, y_train, **fit_params)
  File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\base.py", lin
e 1336, in wrapper
    return fit_method(estimator, *args, **kwargs)
    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\pipeline.py",
line 621, in fit
    self._final_estimator.fit(Xt, y, **last_step_params["fit"])
  File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\base.py", lin
e 1336, in wrapper
    return fit_method(estimator, *args, **kwargs)
    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\discriminant_
analysis.py", line 1132, in fit
    raise linalg.LinAlgError(
numpy.linalg.LinAlgError: The covariance matrix of class 1 is not full rank. Increas
e the value of `reg_param` to reduce the collinearity.
```

```
-----
1 fits failed with the following error:
Traceback (most recent call last):
  File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\model_selecti
on_validation.py", line 833, in _fit_and_score
    estimator.fit(X_train, y_train, **fit_params)
  File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\base.py", lin
e 1336, in wrapper
    return fit_method(estimator, *args, **kwargs)
    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\pipeline.py",
line 621, in fit
    self._final_estimator.fit(Xt, y, **last_step_params["fit"])
  File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\base.py", lin
e 1336, in wrapper
    return fit_method(estimator, *args, **kwargs)
    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\discriminant_
analysis.py", line 1132, in fit
    raise linalg.LinAlgError(
numpy.linalg.LinAlgError: The covariance matrix of class 0 is not full rank. Increas
e the value of `reg_param` to reduce the collinearity.
```

Out[11]: ▾ AutoSklearnClassifier ⓘ

► Parameters

```
In [12]: print("\n=== Cross-Validation Summary ===")
print(f"Mean ROC-AUC: {np.mean(cv_roc_auc):.4f}")
print(f"ROC-AUC Std: {np.std(cv_roc_auc):.4f}")
print(f"Mean F1: {np.mean(cv_f1):.4f}")
```

```
=== Cross-Validation Summary ===
Mean ROC-AUC: 0.9779
ROC-AUC Std: 0.0059
Mean F1: 0.9509
```

```
In [13]: print("\n=== Selected Model (AutoSklearn2) ===")
print(automl_final.best_params)
```

```
=== Selected Model (AutoSklearn2) ===
{'preprocessor': 'standard_scaler', 'classifier': 'random_forest'}
```

```
In [20]: from auto_sklearn2 import AutoSklearnRegressor

auto_sklearn_full = AutoSklearnRegressor(time_limit=120)
auto_sklearn_full.fit(X_encoded, y)

# Try to report best params if available
best_params = getattr(auto_sklearn_full, "best_params", None)
if best_params is None:
    best_params = getattr(auto_sklearn_full, "best_params_", "N/A")

print("\n=== Final Model Trained on Full Data ===")
print(f"Best params: {best_params}")
```


Model Selection Statement

Model Selection Decision

Using AutoSklearn2 with cross-validation, the framework selected a Random Forest classifier as the best-performing model for this task.

AutoSklearn2 identified Random Forest as the best-performing model. In the next step, I explored both Random Forest and Logistic Regression. While Random Forest offered higher predictive power, Logistic Regression provided better interpretability, which is important for risk-screening contexts.

In []: